



One maintainer. One open-source repo. Aligned with 23 security & privacy standards.

We did *not* write 23 checklists.

How good architecture — not a pile of paperwork — keeps a small open-source project audit-ready across the world's privacy and security rules, with no compliance team and no audit budget.

↳ every slide cites a file • automated tests fail the build if a claim stops being true

The standards wall

23 frameworks · 6 domains · across the US, EU, UK & global bodies.

Domain	Where it applies	Frameworks aligned to
Information security	Global · US · UK	ISO 27001 · SOC 2 Type II · NIST CSF 2.0 · FedRAMP · BSIMM · UK Cyber Essentials · NCSC CAF
Privacy & data rights	EU · UK · US · APAC	GDPR / UK GDPR · Global CBPR
Healthcare data	US	HIPAA · HITRUST CSF
AI & agentic systems	EU · US · Global	EU AI Act · NIST AI RMF · OWASP LLM Top 10 · OWASP Agentic Top 10
AI-tool (MCP) security	Global · US	OWASP MCP · CSA MCP · NSA MCP Guidance
Supply chain & protocol	EU · Global	EU CRA · NIS2 · FIRST PSIRT · MITRE ATT&CK · RFC 9700/9449

No compliance team. No budget for 23 audits. So how?



The trick: 23 standards → 8 properties

Strip the labels, and they demand **the same handful of things**:

Access control

Encryption rest + transit

Audit trail

Data minimization

Tenant isolation

Supply-chain integrity

Vulnerability handling

Transparency

Compliance teams call this *control mapping* — and it's usually a 1,400-row spreadsheet and a consultant. **We built the 8 properties into the code instead.**

The payoff: add the 23rd standard and you write zero new code — you already satisfy it. Compliance cost stops scaling with the number of regimes.



The architecture that satisfies them

The 8 properties are baked into *how the code is built* — so they hold everywhere, not just where someone remembered to add them:

- **One way in for every dependency** — nothing reaches in through a back door, so there's one place to audit
- **Swappable parts behind clean seams** — the cache, search, and audit layers can change without rewriting callers
- **One encryption routine** — AES-256 everywhere data is stored, never re-implemented per feature
- **One safe web-fetch client** — *every* outbound request goes through the same SSRF-checked path
- **An audit log on every tool call** — structured, secrets stripped out, and it never slows the request

"Compliance through architecture, not bolt-on checklists."



The safest config is the *default* config

Run it the normal way — locally, inside your AI app — and there's **no network port open, no login to misconfigure, nothing exposed to the internet**. The app that launched it is the only thing that can reach it. The typical user *can't* hold it wrong, because there's nothing to set.

The heavyweight controls for shared, multi-user deployments — sign-in, per-customer isolation, rate limits, encryption, audit logs — only switch on when an operator deliberately runs it as a network server.

Secure by default, permissive by configuration. Power is unlocked by an *explicit* choice, never the reverse.



AI tools have brand-new threat classes

Give an AI a tool that fetches any web page for it, and you've created two attack surfaces ordinary web-security checklists miss:

It can be tricked into attacking your own network (*SSRF*). So before any fetch we check every resolved address against 19 blocked ranges + cloud credential endpoints, connect only to that exact address, and re-check on every redirect — a hijacked link can't reach inside.

The pages it reads are untrusted input — a booby-trapped page can try to hijack the AI (*prompt injection*). So scraped text is cleaned, size-capped, and clearly flagged to the model as "data, not instructions."

Why now: prompt injection is the **#1 risk** on OWASP's LLM list, and the major AI standards don't yet cover *agentic* tools like this one — making it a live case study for a gap regulators are still writing.



An erasure registry you can't outrun

Privacy starts with collecting little: the cache is content-addressed (not keyed to a user), and the operator — not us — is the data controller. But the moment a feature *does* store personal data, "right to be forgotten" has to actually work.

So it's enforced structurally: **every store that holds personal data must register an Exporter + Eraser**. One `(tenant, user)` request fans out to all of them — and each store ships a round-trip **release-gate test**.

Add a new feature and forget to wire its eraser? CI fails — not an auditor. GDPR access / portability / erasure (Art. 15 / 17 / 20) becomes a property of the build, not a promise in a policy PDF.

shipped in v1.13



Consent as a primitive: record → verify → honor

The AI-tool standard (MCP) says *asking* the user for consent is the **client app's** job — so most servers do nothing. But whoever *stores* the data is, in law, the **data controller**, and GDPR / Quebec Law 25 make *them* prove consent was given and honor a withdrawal — a duty a login token can't discharge.

So the server treats consent as three things it actually does:

- **Records** it — encrypted, logged, with a typed purpose ("memory," "analytics," "workspace").
- **Verifies** it on every access — no record, no processing (it defaults to *off*).
- **Honors** it — a withdrawal automatically erases the data it covered.

shipped in v1.13



The docs are tested, not trusted

"Keep the docs accurate" isn't a good intention here — it's enforced by a stack of small mechanisms, so drift is hard to write and impossible to merge quietly:

Layer	What keeps it honest
Code is the source of truth	Docs never hardcode tool lists, counts, versions, or env tables — they point at the file that defines them, so there's nothing to fall out of sync
Rules the AI writes by	The agent guide (CLAUDE.md) makes "every claim links to its file" and "no duplicated facts" mechanical rules for Claude / Copilot / Cursor — the way most code now lands
A test reads the docs	At build time it parses docs/T00LS.md , starts a real server, and fails the build if the documented tools, output shapes, or read/write flags don't match reality
Gates that can't be skipped	A pre-commit hook catches it locally; CI re-runs the drift check on every PR — even docs-only ones, so a doc edit alone can't slip past

The result: the claims in these docs can't quietly stop being true. *Read the code, not the marketing* — because here, the docs answer to the code.



So what is this actually worth?

The same architecture pays off differently for everyone who touches it:

If you're...	What you get
A user	The safe setup is the <i>default</i> — nothing to configure, nothing to leak
An operator / buyer	One small binary you can take to a hospital, an EU regulator, or a federal review — evidence links to code, not slideware
A developer / contributor	Add a feature and the guardrails (consent, erasure, audit, tests) come <i>with</i> it — you can't ship the unsafe version
A founder / eng leader	Compliance cost scales with <i>features you turn on</i> , not with the number of regulations or headcount

Compliance stops being a project you fund and becomes a property you inherit.



What transfers to any project

1. **Map standards to primitives, not checklists** — satisfy many at once.
2. **Make the default the safe one** — gate power behind explicit config.
3. **Let compliance scale with features** (tiers) — not a big-bang program.
4. **Encode the rules as constraints, enforce them in CI** — including for AI coding agents.
5. **If a doc can be wrong without a test failing, it will be** — so test it.

Honest boundaries: "aligned with," not "certified." The local and server threat models differ. By default it stores little; the features that do store personal data are opt-in, consent-gated, and erasable — not absent.



Read the code, not the marketing.

Every slide cited a file. Tests fail the build if those claims stop being true.

The receipts: `v1.10` tenant isolation → `v1.11` zero-config fallback → `v1.12` HTTP hardening → `v1.13` consent + GDPR erasure.

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